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CENTRO DE EMPREGO E FORMAÇÃO PROFISSIONAL DE BRAGA

Technical Report – Overheat Detector

T3 - Specialist Technician in Automation,
Robotics, and Industrial Control

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Temperature Sensor (LM35)

The LM35 is a precision temperature sensor that converts temperature into a proportional voltage output. In this circuit, its purpose is to provide a voltage that rises as the temperature increase, allowing the comparator to detect when a preset temperature threshold is exceeded. Because its output changes linearly with temperature, the LM35 offers a stable and accurate way to monitor temperature and trigger the alarm when the system becomes too hot.

Comparator Reasoning (CA3130)

The CA3130 operates as a comparator, comparing the voltage from the LM35 with a reference voltage. The non-inverting input + receives the LM35 output, while the inverting input – receives the reference generated by the R3-P1 network. The comparator switches its output HIGH when the LM35 voltage rises above the reference level, indicating that the trip temperature has been reached. The R3-P1 network forms an adjustable voltage divider. By varying P1, the reference voltage is set, which defines the temperature at which the circuit triggers.

Two-Transistor Buffer (T1–T2)

Two NPN stages are used to make sure the 555 receives a clean and reliable trigger. The first transistor strengthens the weak output signal from the CA3130, and the second improves isolation and removes noise. Together, they ensure a sharp, stable LOW trigger for the 555.

555 Astable Tone

A piezo buzzer requires an alternating signal to produce sound, because a steady DC output cannot generate a continuous audible alarm. The human ear detects vibrations from an AC signal, making the tone audible, while a steady DC produces no sound. Piezo elements operate more efficiently with alternating signals, consuming less power, and an oscillating tone is more noticeable, clearly indicating an alarm condition. In this circuit, the 555 timer in astable mode generates the required frequency, typically in the kHz range, to drive the buzzer effectively.



Motor (fan)

In the second part, a new circuit was added to control a motor (fan) using a relay (LU5R). Calculations were performed to determine the required base current for the transistor (I_β), which allowed proper selection of the base resistor to ensure reliable operation of the relay and moto.

$$\beta = 10\% \text{ de saturação}$$

$$I_c = 40mA$$

Calculate the base current I_β using the transistor's current gain:

$$\beta = \frac{I_c}{I_\beta}$$

$$I_\beta = \frac{I_c}{\beta}$$

$$I_\beta = \frac{40}{10}$$

$$I_\beta = 0,004A$$

R_β is calculated using the supply voltage V_{Tp1} and the base-emitter voltage V_{be} :

$$V_{be} = 800mV$$

$$V_{Tp1} = 3,1V$$

$$R_\beta = \frac{V_{Tp1} - V_{be}}{I_\beta}$$

$$R_\beta = \frac{3,1 - 0,8}{0,004}$$

$$R_\beta = 580\Omega$$

In conclusion, the addition of the relay-controlled motor circuit was successfully implemented. Calculations for the transistor base current and proper selection of the base resistor ensured reliable switching, allowing the motor to operate safely and effectively as part of the system.